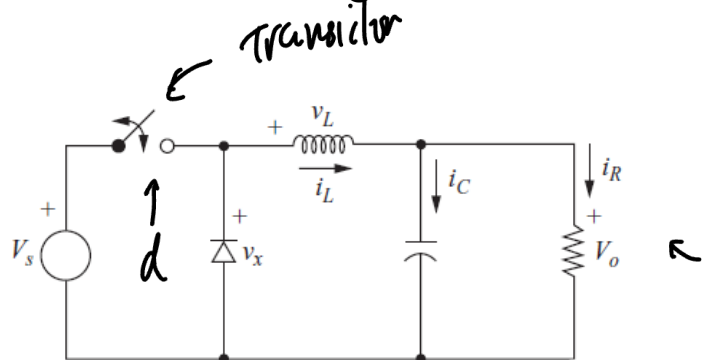


Modelagem do Conversor CC-CC Buck e Diagrama de Blocos do Controle

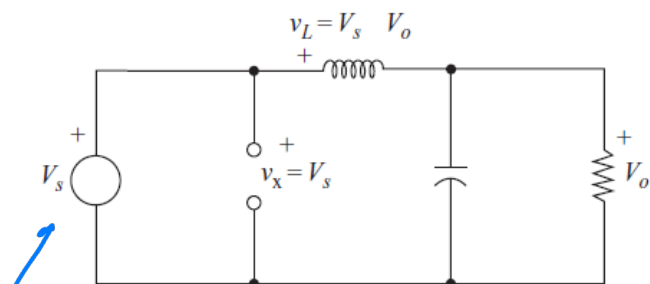


(a)

FT estática

$$V_o(d, R, L, f, \dots)$$

$$V_o =$$

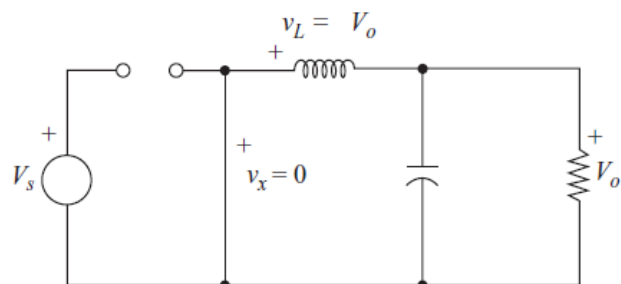


(b)

$$v_L = V_s - V_o$$

$$V_o = V_s \cdot D$$

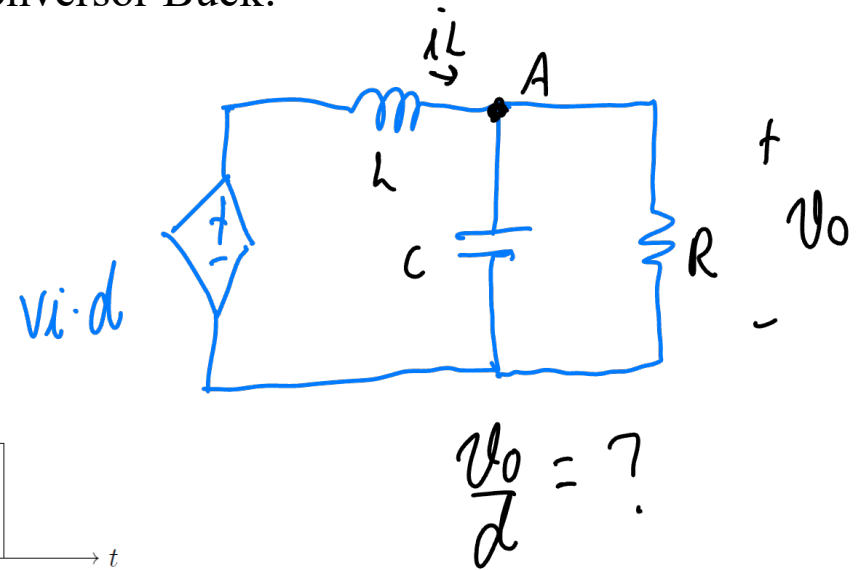
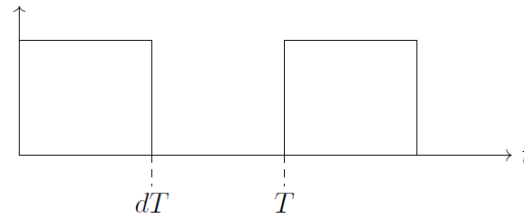
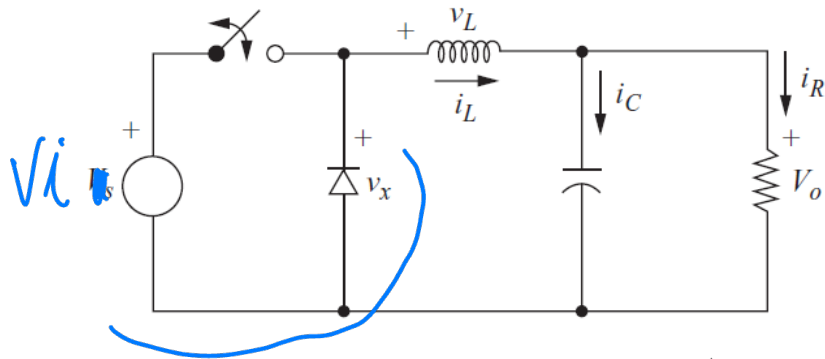
Ponto de operação



(c)

$$v_L = -V_o$$

Obtenção da Função de Transferência dinâmica do Conversor Buck:



$$\rightarrow i_L(t) = i_C(t) + i_o(t)$$

$$\rightarrow i_o(t) = \frac{v_o(t)}{R}$$

$$\rightarrow i_C(t) = C \frac{dv_o(t)}{dt}$$

$$\hookrightarrow v_L(t) = L \frac{di_L(t)}{dt} = V_i d(t) - v_o(t)$$

$$V_i d(t) - v_o(t) = L \frac{d}{dt} \left(C \frac{dv_o(t)}{dt} + \frac{v_o(t)}{R} \right)$$

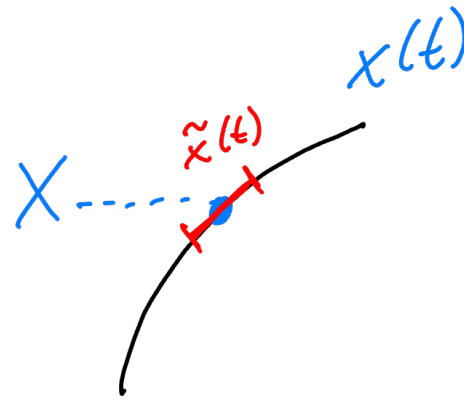
$$V_i d(t) - v_o(t) = LC \frac{d^2 v_o(t)}{dt^2} + \frac{L}{R} \frac{dv_o(t)}{dt}$$

$$\Rightarrow \quad V_i d(t) = LC \frac{d^2 v_o(t)}{dt^2} + \frac{L}{R} \frac{dv_o(t)}{dt} + v_o(t) \quad \mathcal{N}_{io} \text{ linear}$$

$\uparrow$

$\uparrow$

Linearização num ponto de operação:



$$x(t) = X + \tilde{x}(t)$$

$$\rightarrow d(t) = D + \tilde{d}(t)$$

$$\rightarrow v_o(t) = V_o + \tilde{v}_o(t)$$

$$V_i(D + \tilde{d}(t)) = LC \frac{d^2}{dt^2} (V_o + \tilde{v}_o(t)) + \frac{L}{R} \frac{d}{dt} (V_o + \tilde{v}_o(t)) + V_o + \tilde{v}_o(t)$$

$$\Rightarrow \underbrace{V_i D + V_i \tilde{d}(t)} = LC \frac{d^2 \tilde{v}_o(t)}{dt^2} + \frac{L}{R} \frac{d \tilde{v}_o(t)}{dt} + \underbrace{V_o} + \tilde{v}_o(t)$$

↳ separando valores fixos e pequenos sinais

$$\underline{V_i D = V_o} \quad \swarrow$$

$$V_i \tilde{d}(t) = LC \frac{d^2 \tilde{v}_o(t)}{dt^2} + \frac{L}{R} \frac{d\tilde{v}_o(t)}{dt} + \tilde{v}_o(t) \quad \swarrow$$

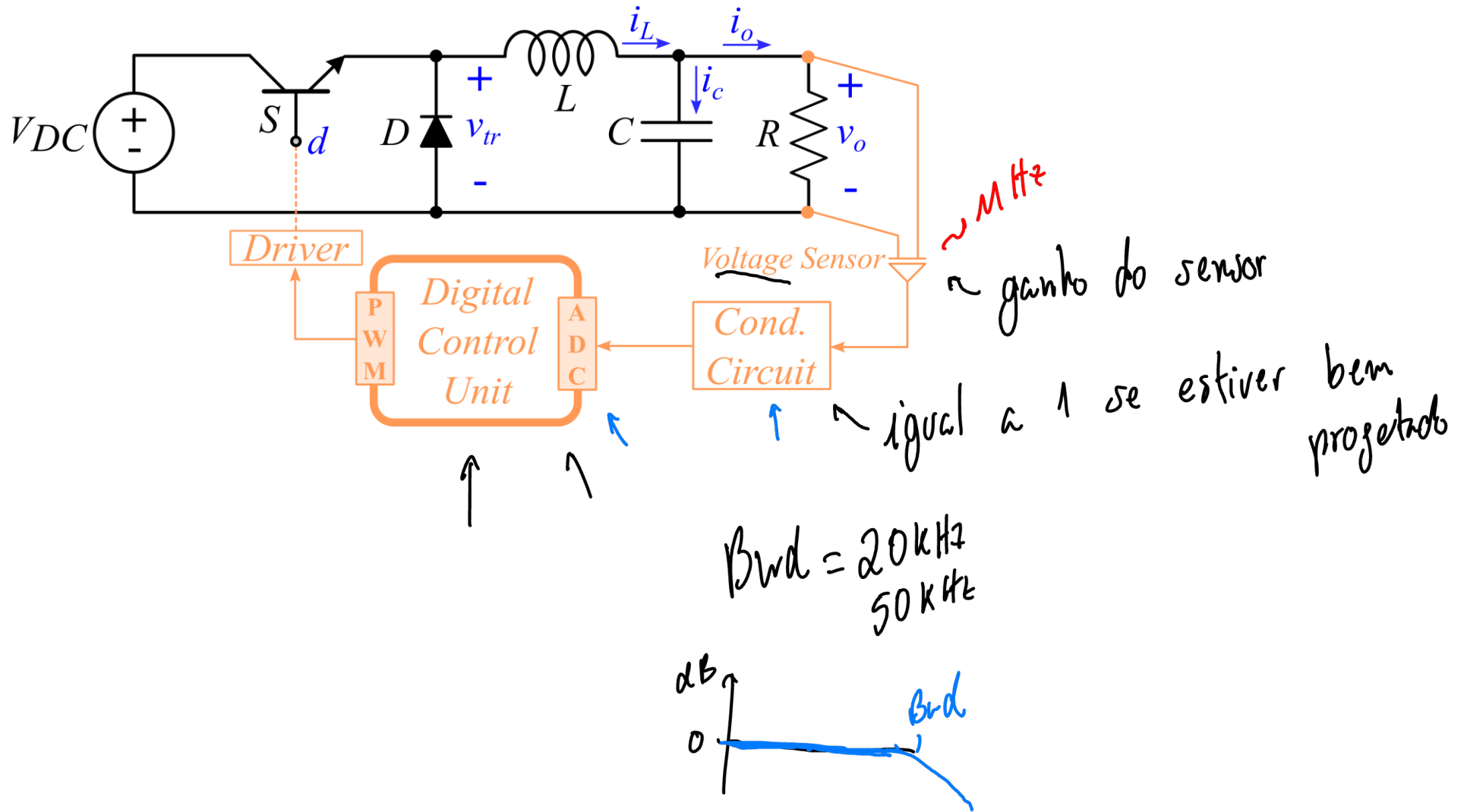
Laplace

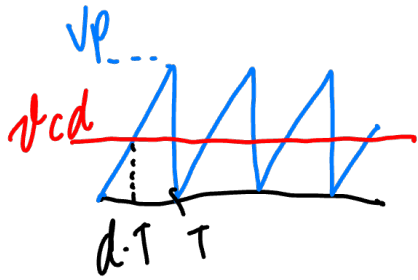
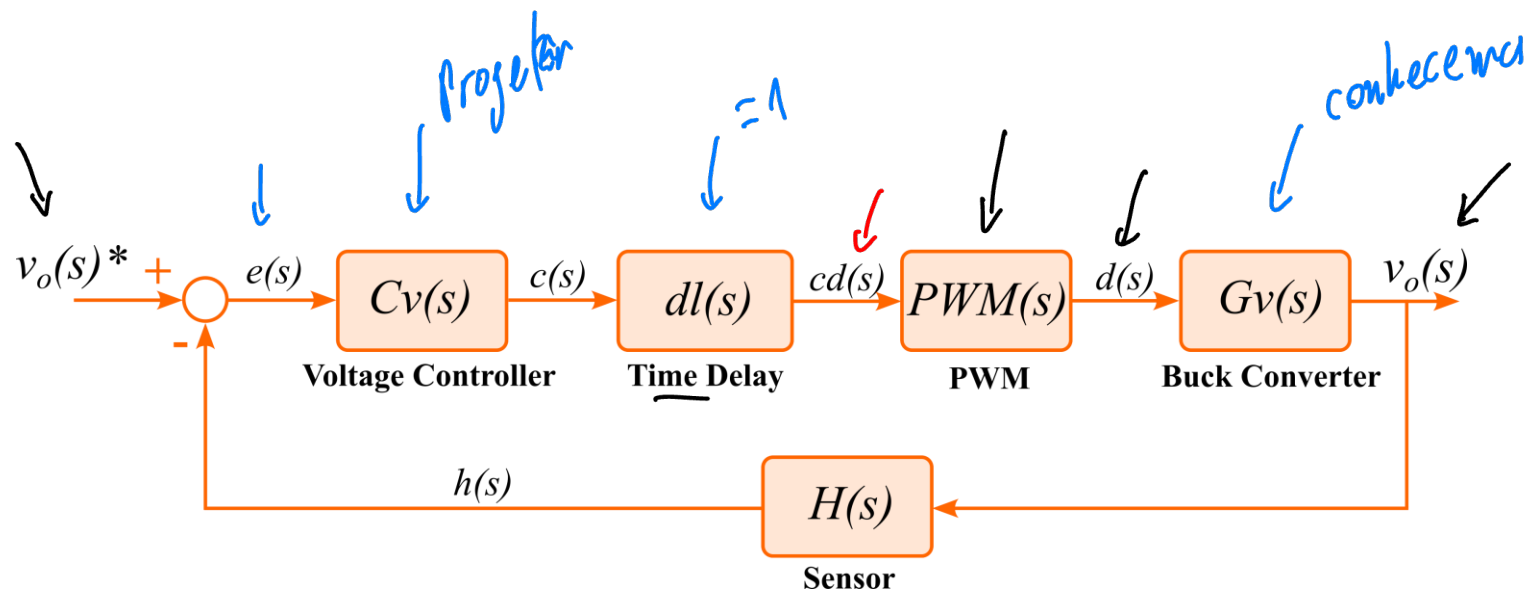
$$V_i \tilde{d}(s) = \tilde{v}_o(s) \left(LCs^2 + \frac{L}{R}s + 1 \right).$$

$$\boxed{\frac{\tilde{v}_o(s)}{\tilde{d}(s)} = \frac{V_i}{LCs^2 + \frac{L}{R}s + 1} = G_v(s)}$$

← Não depende
do ponto de operação
 D, V_i

Diagrama de Blocos:





$$PWM(s) = \frac{d(s)}{cd(s)}$$

$$\frac{v_{cd}}{v_p} = \frac{d \cdot T}{T}$$

$$\frac{d}{v_{cd}} = \frac{1}{v_p}$$

Resumindo:

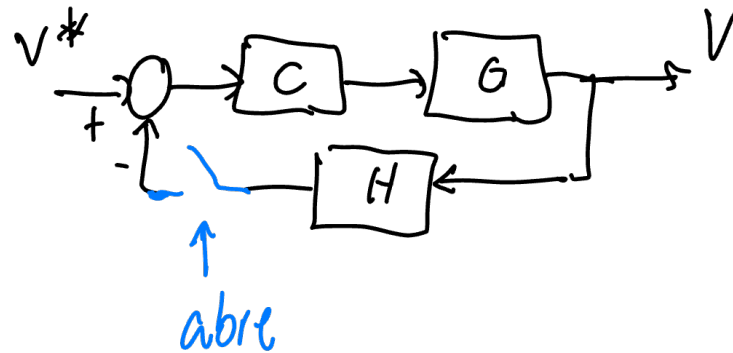
$$G_v(s) = \frac{V_i}{LCs^2 + \frac{L}{R}s + 1}$$

$$P_{WM}(s) = \frac{1}{V_p}$$

$$d_l(s) = 1$$

$$H(s) = H_v \text{ (ganho do sensor)}$$

$$C_v(s) \Rightarrow \text{será projetado}$$



$$FTMA_u(s) = G \cdot H$$

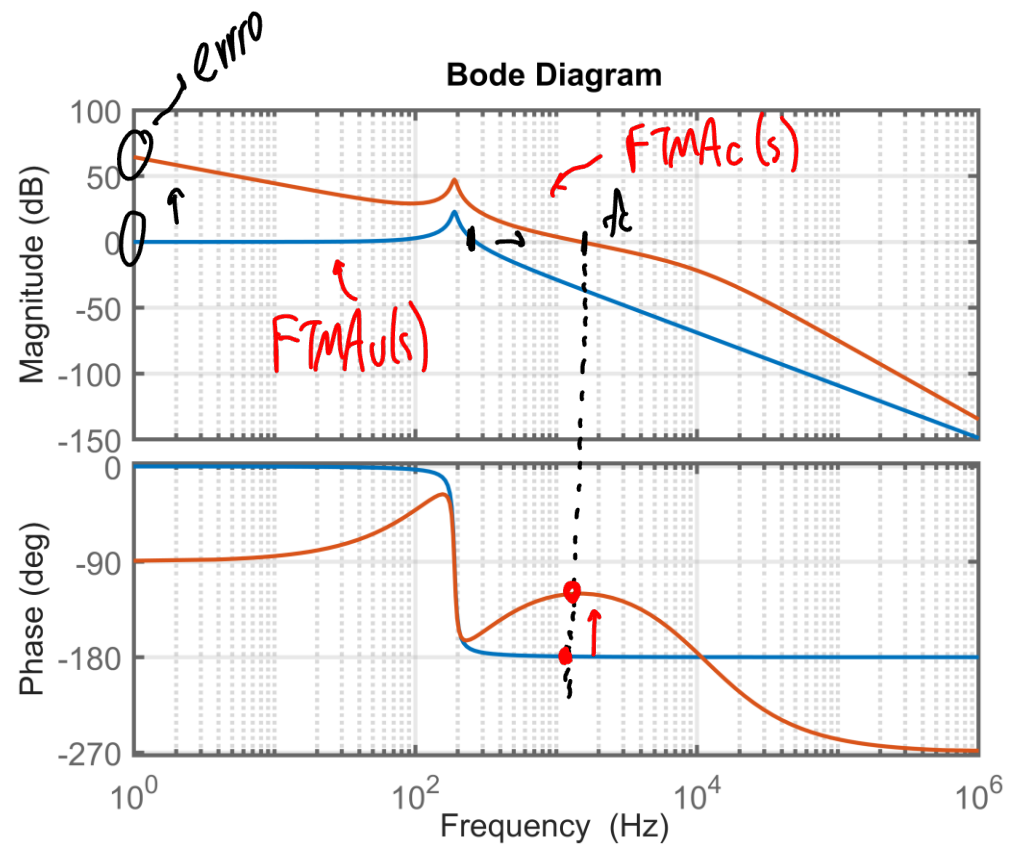
$$FTMA_c(s) = C \cdot G \cdot H$$

$$FTMA_u(s) = P_{WM}(s) G_v(s) \cdot H(s)$$

$$FTMA_c(s) = C(s) \cdot P_{WM}(s) G_v(s) \cdot H(s)$$

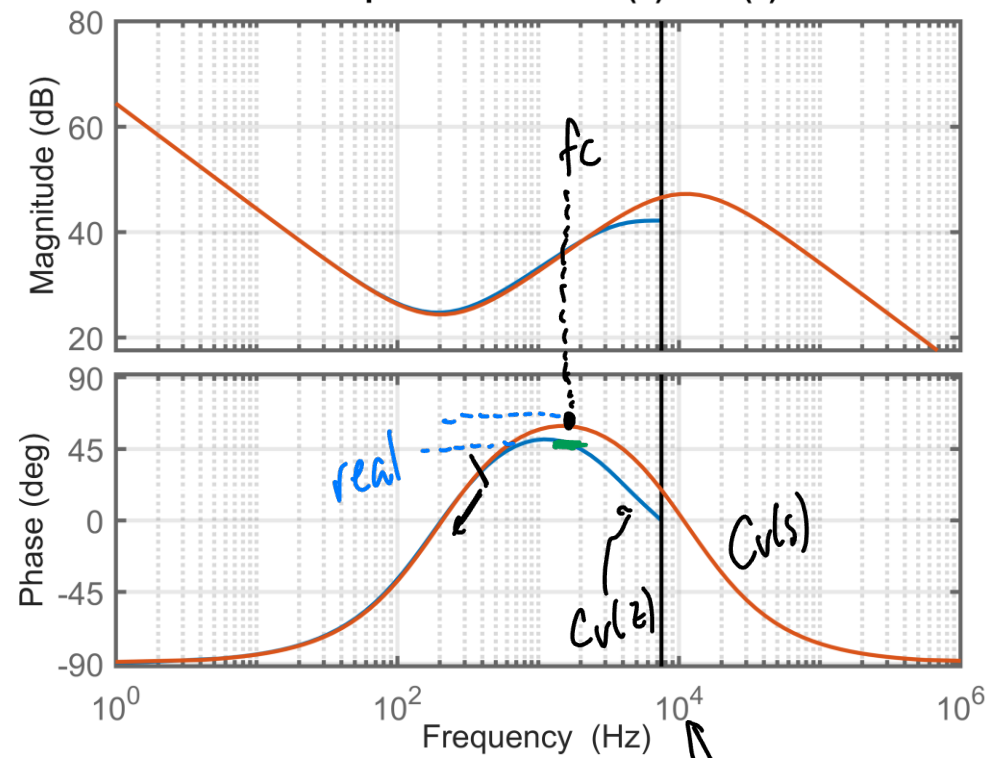
Projeto pelo método
do Fator K

Exemplo de diagramas de bode de FTMAu(s) e FTMAc(s):



S

Comparativo entre $C_v(s)$ e $C_v(z)$



Bode em Malha fechada

$0\text{dB} = 1$

